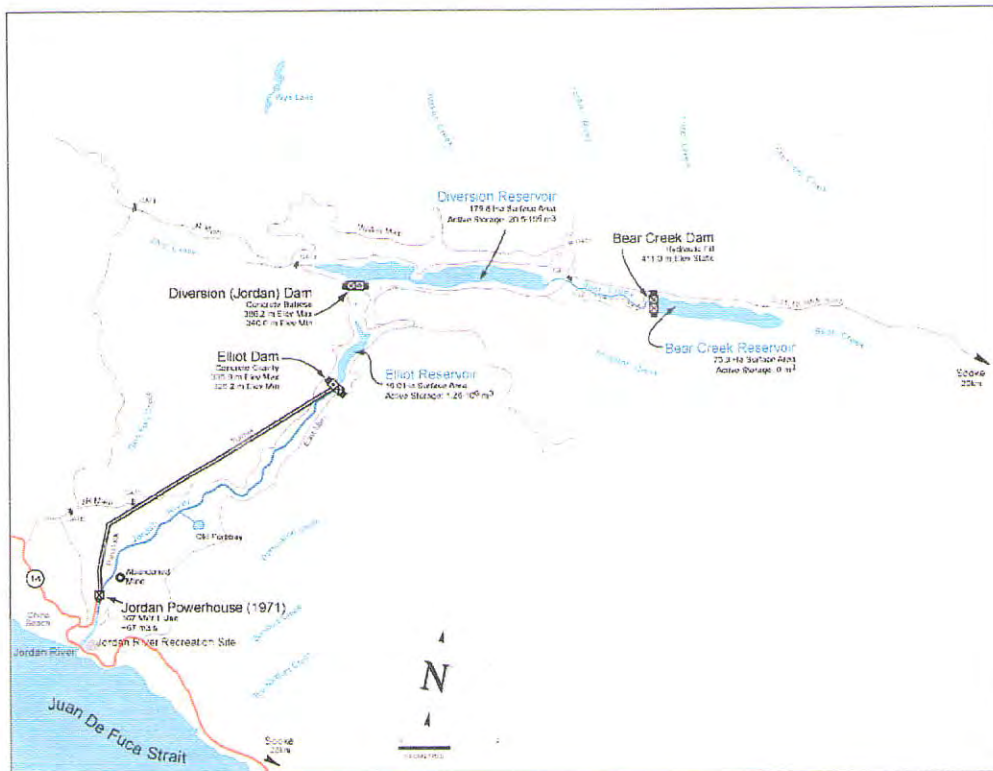


DARRIN SHERBOT
41556

JORDAN RIVER HISTORICAL SALMONID SURVEY

*Anecdotal Information Regarding Salmonid Presence or Absence
on the Jordan River prior to 1971*



Prepared for:
BC HYDRO

By:
RRL Recreation Resources Ltd.
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Executive Summary

BC Hydro (BCH) is currently developing a Water Use Plan (WUP) for the Jordan River (JR) operations in consultation with government and non-government representatives for environment, social, economic and First Nations issues.

In order to address fishery information data gaps, a survey of anecdotal information from local sport fishers, First Nations, and other knowledgeable residents was initiated to determine the historical presence and distribution of salmonids in the lower JR. In particular, to establish whether, prior to 1971, steelhead were ever present above a series of cascades and falls which are located approximately 2 km upstream of the river mouth.

BCH contracted an independent consulting company, RRL Recreation Resources Ltd. to undertake the survey. The survey data will be used to evaluate the necessity of other potential fishery studies.

BC Hydro has operated powerplants on the lower JR since 1912. The watershed has been actively logged since the 1880's. Sites between Elliot dam and the current powerhouse have been mined intermittently for copper since the 1900's. Mining operations were suspended in the 1980's.

Historic fishery information prior to the anecdotal survey, weighted with the current presence of rainbow trout in the boulder-cascade channel of the lower JR, suggests that a steelhead run may have been supported under certain flow years. There is no anadromous fish production within the lower JR at present. Salmonids were not seen after the construction of the new powerhouse (1969 – 1971) and the decommissioning of the old tailrace.

Interviewees were chosen based on their knowledge of the JR prior to the 1970's. Further names were provided by the interviewees contacted, by long time businesses in Jordan River, by members of Sooke and Victoria fishing clubs, by government fisheries biologists, and by other individuals knowledgeable of the JR area or fishing areas on southern Vancouver Island. The appendices contain the detailed interview information.

Results

Thirty-nine individuals or organizations were contacted. Of these, eighteen had information specifically about the JR and twelve were quite familiar with the river.

Ten of the twelve contacts familiar with the JR provided similar reports regarding the fish species, locations, and quantity, quality and the time of year. Two of the twelve contacts provided considerably different reports.

The fish species present in the JR prior to the 1970s were pink, chum and coho salmon, sea-run cutthroat trout, steelhead, resident trout (rainbow and cutthroat), perch, smelt, and bulltrout. Salmon were present in the JR in the fall from September until November. There were some discrepancies in regards to when steelhead were present; some said in the fall, while others said March until May or June.

Salmon, cutthroat trout and steelhead were reported as being in two major areas, the old BC Hydro tailrace, and the JR itself, with the majority in the tailrace. This was a significant spawning area and it was filled with fish during the fall spawning time. In the JR itself there was a spawning area approximately 1.6km up the river, before the first major waterfall, by the old mine. Salmon (principally pinks) and steelhead would spawn in this area.

The species with the largest run was pink salmon with over 1000 fish per year. Coho salmon numbered 500 - 1000 per year and weighed between 3 - 8 lbs.

Some steelhead were present in the old tailrace; however the majority were in the main river. The steelhead would travel approximately 1.6km up the river where they would spawn in the pool by the mine, below the first cascade. With the exception of two individuals, the contacts familiar with the river, and with the area above the waterfall, were not aware of steelhead in the portion above the falls. In addition, many specifically mentioned that the steelhead spawned at the bottom of the falls and were not be able to get past the first waterfall.

No clear indication of the size of the steelhead run was given. In addition, the reports of fish size varied between 4 lbs. to 20 lbs., with the majority suggesting that 6-8 lbs. was about average.

There was a fairly healthy trout population from the mouth of JR up into the reservoirs as far as Bear Creek and Diversion, in addition to some trout being present in the old tailrace. Trout were present in the area above the cascades, in the canyon, and in the reservoirs. Both rainbow and cutthroat trout were present, some of the cutthroat being sea-run. An average trout was said to be approximately 6-12 inches long.

Smelt, perch, herring and bullheads were present in the JR or at the mouth of the river prior to 1971.

Two individuals provided information which was contrary to what was stated by the other interviewees. One individual gave conflicting information in two interviews about a series of fish ladders up the JR in the canyon area. However; in both interviews he stated that he observed salmon and steelhead in the JR from the river mouth to 8km up river, the majority of the fish stopping at 4.8km or 6.4km up from the river mouth. No one else had seen fish ladders.

A second contact, who was interviewed in-person by a representative from the Ministry of Environment, indicated that there were steelhead in the reach upstream of the "obstruction". He accessed the canyon areas via a wooden ladder.

Many of the same names were suggested by interviewees. It became apparent that only a few people either lived at JR or are familiar with the river. It is difficult to determine how many more people not currently on the contact list would have information regarding fish species on the Jordan River prior to 1971.

Additional interviews could be undertaken in an effort to clarify the information by two contacts who indicated that salmonids were present above the first cascades in the Jordan River. However, the number of individuals that may have such information appears to be limited, and the majority of interviews that were conducted suggest that salmonid presence was only in the area below the first cascades in the Jordan River.

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1.0 Project Background

BC Hydro (BCH) is currently developing a Water Use Plan (WUP) for the Jordan River operations in consultation with government and non-government representatives for environment, social, economic and First Nations issues.

In partial fulfillment of addressing fishery information data gaps, a survey of anecdotal information from local sport fishers, First Nations, and other knowledgeable residents was initiated to determine the historical presence and distribution of salmonids in the lower Jordan River. In particular, the Consultative Committee (CC) was interested in establishing whether steelhead were ever present above a series of cascades and falls which are located approximately 2 km upstream of the river mouth prior to 1971.

BCH contracted an independent consulting company, RRL Recreation Resources Ltd. (RRL) to undertake the survey. RRL is located on southern Vancouver Island, and since 1986, has specialized in the inventory and analysis of outdoor recreation resources throughout British Columbia.

The objective of this survey work was to:

- *Catalogue anecdotal information on the historical presence or absence of salmonids (and other fish) endemic to the Jordan River prior to 1971.*
- *Determine salmonid spawner distribution along the Jordan River and whether species were observed above the first cascade/fall series ~ 2.1 km upstream from the river mouth*
- *Annotate fish numbers, location, condition, species, and spawning timing*

The survey data will be used by the Jordan River Fisheries Technical Committee (FTC) and the (CC) to evaluate the necessity of other potential fishery studies.

1.1 Hydroelectric Operations History

The original power development was initiated in 1909 and completed in 1912 by Victoria Light & Power Company. (Conway 1915). The powerplant operated continuously between 1912 and 1971, producing approximately 26.4 megawatts of electricity until replaced by the present system on the west bank. Four Pelton type generation units were serviced by four penstocks from the original forebay. (Conway 1915)

Forebay water levels were maintained by an 8.5 km long wooden flume from Diversion Reservoir. Diversion Reservoir does not actually receive any water "diverted" from another watershed. Diversion Dam and Diversion Reservoir were named because it was used to divert the 1912 project river flows into the old forebay. The wooden flume was rebuilt twice, once in 1928 and again in 1948. (BCH. Draft 2000 JOR Fisheries Overview).

Victoria Light & Power created a man-made channel from the old power house to the river in 1915. (WFP 1997) The channel was approximately 500 m in length and served as a significant spawning area for pink and chum salmon. (DFO 2000, BCH 1955).

In 1932, a 2 MW vertical shaft Francis turbine was installed in the bay between buttresses on Diversion Dam. This plant was decommissioned in 1962. (BCH 1977B).

Under the redevelopment program (1969 – 1971) the dam at Diversion Reservoir was improved, the Elliot dam was created to replace the existing east forebay and a single high head Francis turbine (167 MW) was installed in the new powerhouse on the east bank. The powerhouse was fed by a new pressure tunnel and penstock (Total Length: 7.2 km) from Elliot dam, bypassing the original forebay. (BCH Draft 2000, JOR Fisheries Overview).

1.2 Forestry Operations

The watershed has been actively logged since the 1880's. Western Forest Products (WFP) have managed logging operations and conducted booming and towing operations in the lower Jordan River since 1934. (BCH Draft 2000, JOR Fisheries Overview).

The gravel bar at the mouth of the river has been frequently dredged to facilitate log boom towing. During some operations this occurred annually. (WFP 1997).

1.3 Mining

Sites between Elliot dam and the current powerhouse have been mined intermittently for copper since the 1900's. Mining operations were finally suspended following the depression of copper prices in the 1980's. (BCH Draft 2000, JOR Fisheries Overview).

Prior to 1960, mine tailings were deposited along the east bank. Operations between 1960 – 1974 were permitted to discharge approximately 450 m³/d of mine tailings to the nearshore shelf just off the mouth of Jordan River (500 m east of the river mouth, and 500 m out). The pipeline failed at least twice during this period, depositing tailings closer to the shoreline. Tailings had the consistency of very fine sand. (US Bureau of Mines 1994).

In 1963, a mine shaft under the Jordan river collapsed ("mine blowout") while the mine was being operated by Cowichan Copper Mine. Additional slumping and sloughing have occurred periodically along the east river bank as abandoned mine shafts become saturated with water. The debris associated with the slide and runoff through the mining tunnels are characterized by high levels of copper, which can be acutely toxic to aquatic life. (Griffith 1986, Aquametrix 1997).

1.4 Fisheries Background

The Department of Fisheries and Oceans (DFO) tracks the presence of salmon in rivers through the Salmon Escapement Data System (SEDS). Species listed in SEDS are noted to spawn in the old BCH tailrace between 1949 – 1970. Note that the accuracy of SEDS records prior to 1970 are uncertain and further decrease in reliability prior to 1950 (DFO M Sheng pers. comm. 13/07/2000).

Table 1 Species Inventory Summary

Species	Notes	Reference	Period	Peak Numbers
Chum O. keta		SEDS	1932 – 1957	1950: 5000 – 10000
Coho O. kisutch		SEDS	1932 – 1965	1950: 1000 – 2000
steelhead mykiss O. mykiss	Winter Run	SEDS	1932 – 1957	1936: 500 – 1000
pink O. gorbuscha	Even Year	SEDS	1952 – 1970	1962: 5000 – 10000
cutthroat trout O. clarki				
cutthroat trout O. clarki	Stocked	Griffith 1996	1983 – Present	
rainbow trout O. mykiss	Stocked	Griffith 1996	1983 – Present	

1.4.1 Anadromous Fish

Historically, small runs of coho, chum and pink salmon were supported by the lower Jordan River. (Hirst 1991). This conclusion was based on anecdotal evidence and the DFO Annual Stream Survey data for Jordan between 1932 – 1970

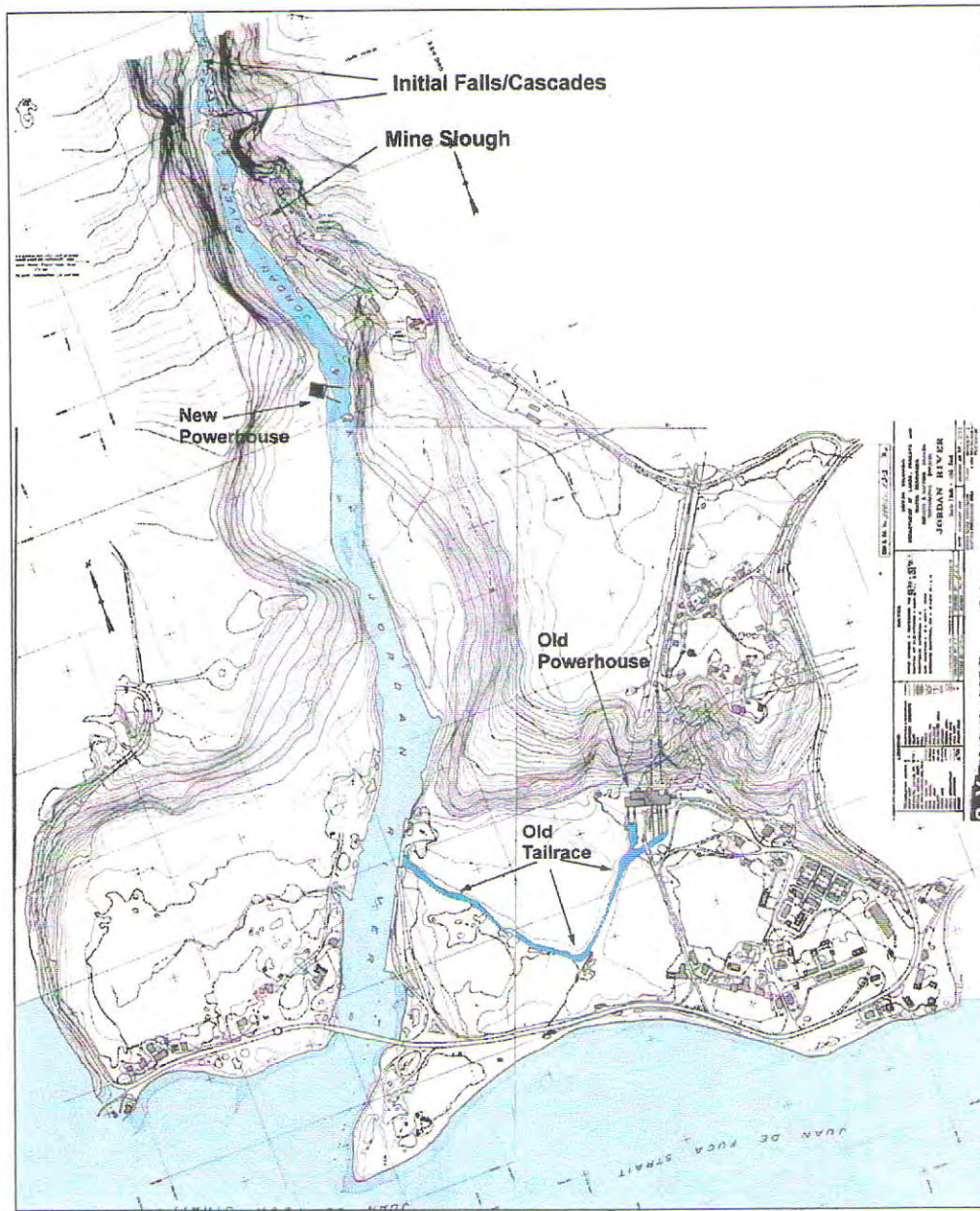
The historic presence of steelhead and sea-run cutthroat has also been noted by DFO records and MELP archives, although Griffith (1996) suggests that sea-run cutthroat may be “better explained by feeding migrations of fish from other systems.” DFO references to steelhead include a winter run only (DFO 2000). Historic information, weighted with the current presence of rainbow trout in the boulder-cascade channel of the lower Jordan, suggests that a steelhead run may have been supported under certain flow years (BCF Ptolemy pers. comm. 07/20/00).

Surveys after 1970 record only the absence of anadromous species. Annotation in the DFO survey and a BCH drawing (544-C11-D307) suggest that the majority of the anadromous spawning occurred in the old powerhouse tailrace (1912- 1971) on the east side of the river. Flows in the 1912 tailrace were maintained by a 35 cfs (~1 m³/s) flow by mutual agreement between BCH and DFO during the years 1964 – 1969. Of note is the 1970 annotation that a pink run that was previously spawning in the old tailrace now spawns in the river proper. Likely this coincided with the decommissioning of the old tailrace during construction of the new powerhouse in 1969 – 1971. (BCH, Draft 2000. JOR Fisheries Overview).

There is no evidence of any anadromous fish production within the lower Jordan River at present (Griffith 1996, BCH Longworth G pers. comm. 2000). Both annotations in the DFO survey records (DFO 2000) and Griffith (1996) indicate that the cessation of salmonid presence is temporal with the construction of the new powerhouse (1969 – 1971) and the decommissioning of the old tailrace.

1.5 Reference Map

Jordan River 1912-1971 powerhouse and tailrace.



2.0 Survey Methodology

2.1 Interviewee Selection

Interviewees knowledgeable of the Jordan River (JR) prior to the 1970's were initially determined through names of individuals provided by Darren Sherbot (BCH) and Marion Lightly (DFO Nanaimo) and suggested by the JOR WUP CC. This list was supplemented by the names of individuals recommended by the interviewees contacted, by long time businesses in JR, by members of Sooke and Victoria fishing clubs, by government fisheries biologists (past and present), and by other individuals suggested to be knowledgeable of the JR area or fishing areas on southern Vancouver Island. See Appendix A for the complete list of individuals contacted. Appendix B contains the individual interview summaries.

The Beecher Bay Band, Nuuchan-nulth Tribal Council, Pacheenalt Nation, Dididaht First Nation and T'sou-ke Nation were also contacted. The T'sou-ke Nation arranged through the Aboriginal Affairs section of BC Hydro for an elders meeting to be held to discuss historical information on the JR. The transcript from this meeting, which was held on December 11, 2000, is included as Appendix C to this report.

The Beecher Bay Band and Nuuchan-nulth Tribal Council indicated that JR was not within their area and suggested we contact the First Nation groups closer to JR.

2.2 Interview Methodology

Telephone interviews to collect information about fish species on the Jordan River (JR) were conducted over a two and a half week period, Nov. 7 – 20, 2000. The interviewers identified themselves as working for RRL Recreation Resources Ltd., a private consulting firm on contract to BC Hydro.

The interviewees were asked the following questions:

- Were they familiar with the JR?
- What species of fish were present on the JR prior to the 1970's?
- Where on the river were the different species of fish found?
- What time of year?
- What years - when was the interviewee living in JR, when were the fish there?
- Size of the fish (for the different species)?
- Numbers of fish (for the different species)?

If the interviewee hadn't mentioned steelhead or the canyon area above the boulders they were asked the following:

- Do they remember steelhead being present on the JR? -if so where?
- Were they familiar with the area "above the boulders"?
- Were there any fish in the area "above the boulders" (canyon area)?
- What type of fish (size, quantity etc.)

Before the interview ended the interviewees were asked if they knew of anyone else who might be knowledgeable about the JR prior to the 1970s and their name and number if possible.

All interviews were conducted over the telephone with the exception of one individual where a report and map were submitted in addition to a telephone interview.

2.3 Interview Results

Thirty-eight individuals or organizations were contacted through telephone interviews. One individual was interviewed in person by Ministry of Environment staff. Of those contacted eighteen had information specifically about the JR, twelve of whom were quite familiar with JR. The other twenty individuals were contacted as it was suggested that they might be familiar with the JR or know of others who were. The following is a summary of the information provided by the interviewees. Appendix A lists the individuals and organizations contacted. Appendix B lists the information from the individual interviews.

Ten of the twelve main contacts with information regarding the JR provided similar reports regarding the fish species, locations, and quantity, quality and the time of year. Two of the twelve contacts provided considerably different reports of the JR. The information from these two contacts is summarized under "2.6 Alternate Results".

2.4 Time Frame

Interviewees provided information on the JR from the 1940s until the early 1970s. All contacts that knew of the mine accident mentioned that there were no fish present in the Jordan River after the mine shaft collapsed. The majority of the people interviewed were familiar with the JR from the 1950's until the mine accident.

2.5 Fish Species

Several species of fish were mentioned as being present on the JR prior to the 1970s including pink, chum, coho, sea-run cutthroat trout, steelhead, resident trout (rainbow and cutthroat), perch, smelt, and bulltrout. The salmon were said to be present in the JR in the fall from September until November. There were some discrepancies in regards to when the steelhead were present; some individuals said that the steelhead were present in the fall, while others mentioned that JR had a winter and a spring run, making the steelhead present in JR from March until May or June.

2.5.1 Salmon

Contacts mentioned that the salmon were present in two major areas, the old BC Hydro tailrace, and the JR itself. The majority of the salmon were said to be present in the tailrace. Salmon species in the tailrace included coho, spring, chum, and pink in addition to the odd cutthroat trout and steelhead. This was a significant spawning area and was filled with fish during the fall spawning time.

In the JR itself the contacts mentioned that there was a spawning area approximately 1 mile (1.6km) up the river, before the first major waterfall, by the old mine. Salmon (principally pinks) would spawn in this area, in addition to steelhead.

Pink Salmon were identified as the largest run, of over 1000 fish a year. Contacts mentioned that there were likely 500 - 1000 coho per year. The salmon were reported to be between 3 - 8 lbs.

2.5.2 Steelhead

Steelhead were also reported present on the JR. Some steelhead were mentioned to be present in the old tailrace; however the majority were said to be in the main river. The steelhead were said to travel approximately one mile (1.6km) up the river where they would spawn in the pool by the mine, below the first cascade. With the exception of two individuals (see section "2.6 Alternate Results") the contacts familiar with the river, and with the area above the waterfall, were not aware of steelhead in the portion above the falls. In addition, many of the individuals specifically mentioned that the steelhead spawned at the bottom of the falls and were not be able to get past the first waterfall.

No clear indication of the size of the steelhead run was given. In addition, the reports of fish size varied between 4 lbs. to 20 lbs., most individuals suggesting that 6-8 lbs. was about average.

2.5.3 Trout

It was indicated that there was a fairly healthy trout population from the mouth of JR up to the reservoirs as far as Bear Creek and Diversion, in addition to some trout being present in the old tailrace. Trout were identified as being present in the area above the cascades, in the canyon, and into the reservoirs. Both rainbow and cutthroat trout were present, some of the cutthroat being sea-run (the cutthroat can be either sea-run or resident). An average trout was said to be approximately 6-12 inches.

2.5.4 Other fish species

Smelt, perch, herring and bullheads were also said to be present in the JR or at the mouth of the river prior to 1971.

Table 2 - Salmonid Presence Summary Table

Species	Location(s)	Months	Number of persons	Estimate of # of Fish	Estimate of size
Salmon: coho, spring, chum, pink, cutthroat	Old BCH Tailrace and Jordan River. Spawning at 1.6km . and in tailrace. Present from mouth before first barrier and in tailrace	August - November	10	Coho 500-1000 Pink 1000	Coho 3-8lbs Chum/pink 15-20lbs
Steelhead	Some in tailrace, most in Jordan River up to 1.6km	March – June	10	No clear indication	4-20lbs (most reporting 6-8 lbs.)
Steelhead	Above barrier at 1.6 km	March – April	2	Caught 2-3 per trip	20lbs
Trout	From river mouth up into reservoirs including in canyon. Some in tailrace	Year around	11	No clear indication-	6 – 12 inches
Smelt, perch, herring and bullheads	River mouth	July August	4	No clear indication	No clear indication

2.6 Alternate Results

Two individuals provided information about the JR, which was contrary to what was stated by the other interviewees. One individual stated in the first interview, that there were a series of fish ladders up the JR in the canyon area. In a follow-up interview, he stated that there were no fish ladders. However, he confirmed his comments from the first interview, observing that there were salmon (unsure of the species) and steelhead in the JR from the river mouth to five miles (8km) up river, the majority of the fish stopping at three (4.8km) or four (6.4km) miles up from the river mouth. This individual lived at JR from 1943 until 1956 when he left at age twelve. This individual's eldest brother was also contacted. He did not mention fish ladders on the JR. In contrast he stated that he was not aware of steelhead in the "canyon area". Other individuals, when asked about fish ladders, stated that there were never any present on the JR.

A second contact, who was interviewed in-person by a representative from the Ministry of Environment, indicated that there were steelhead in the reach upstream of the "obstruction". He accessed the canyon areas via a wooden ladder. The ladder's location and access was adjacent to the Forebay Reservoir in Reach 2 with access from the East side. The interviewer characterized the contacts' recollections as being clear and concise.

3.0 Summary

Later interviews provided names of individuals who grew up in, and/or were familiar with the Jordan River. However, many of the individuals suggested have already been contacted. Therefore it is difficult to determine how many more people not currently on the contact list would have information regarding fish species on the Jordan River prior to 1971.

Additional interviews could be undertaken in an effort to clarify the information by two contacts who indicated that salmon were present above the first cascades in the Jordan River. However, the number of individuals that may have such information appears to be limited, and the majority of interviews that were conducted suggest that salmonid presence was only in the area below the first cascades in the Jordan River.

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Appendix A List of Individuals and organizations contacted

Last Name	First Name	Phone Number	Contacted	Connection
Banner	Cliff	(250) 743-9383		Past resident of Jordan River
Beecher Bay Band		(250) 478-3535	14/11/00	
Billings	Ray	(250) 746-1425	10/11/00	Fish Hatchery Manager – Duncan
Broadland	Ken	(250) 746-1257	14/11/00	Conservation Officer in Duncan
Burns	Ted	(250) 749-6006	10/11/00	Fisheries Consultant
Carter	Ray	(250) 598-1295	9/11/00	Darren Sherbot
Dick	Bob	(250) 646-2031	15/11/00	Western Forest Products
Elliot	Ted	(604) 436-4023	18/11/00	Past resident of JR
Eves	Les	(250) 642-5969	14/11/00	Sooke Salmon Enhancement
Forest	Pat	(250) 642-3988	14/11/00	Sooke Salmon Enhancement
French	Dennis	(250) 920-5601	8/11/00	Marion Lightly
Hall	Don	(250) 724-5757	15/11/00	Nuu-chan-nulth Tribal Council
Hotel at Jordan River		(250) 646-2244	10/11/00	Long time business of the area
Hull	Mell	(250) 642-5548	14/11/00	Sooke Salmon Enhancement
Lajeunesse	Bob	(250) 646-2856	10/11/00	Spoke to his wife (he had passed away)
Lenfesty	Jack	(250) 652-2495	14/11/00	Retired Conservation Officer
Lightly	David	(250) 642-3957	14/11/00	Fisheries biologist at the Sooke Nation
Lightly	Marion	(250) 756-7392	15/11/00	Darren Sherbot, BC Hydro
McCully	Peter	(250) 474-1620	10/11/00	Goldstream Hatchery
Michelson	Randy	(250) 642-4788	15/11/00	Past resident of JR
Nyberg	Arnie	(250) 475-2490	19/11/00	Past resident of JR
Pednault	Bill	(250) 642-3443	15/11/00	Past resident of JR area
Poirier	Dick	(250) 646-2076	9/11/00	Grew up in JR – Darren Sherbot
Poirier	Ken	(250) 746-9338	16/11/00	Past resident of JR
Poirier	Les	(250) 479-3062	07/01/01	Past resident of JR
Ptolemy	Ron	(250) 356-7054	10/11/00	Rivers Biologist at Prov. Gov't
Reid	George	(250) 751-3228	10/11/00	Head, Fisheries Section, MELP Nanaimo
Soderberg	Stewart	(250) 646-2020	15/11/00	Past resident of JR
Soderberg	Mike	(250) 595-1887	16/11/00	Past resident of JR
Steven	John	(250) 642-4756	15/11/00	Retired Fisheries Officer
Trembley	Maurice	(250) 647-5568	14/11/00	Port Renfrew Hatchery
Varney	Glen	(250) 642-5490	14/11/00	Sooke Salmon Enhancement
Victoria Fish & Game Club		(250) 478-2411	9/11/00	
Wade	Les	(250) 646-2060	13/11/00	Long time resident/business owner
Waters	Rob	(250) 478-6009	13/11/00	Fishing writer for the Times Colonist
Wickheim	Maywell	(250) 642-4420	24/10/00	Long term resident of Sooke
Wightman	Craig	(250) 751-3230	16/11/00	Steelhead expert at MELP
Zaccarelli	Wayne	(250) 382-8502	19/11/00	Victoria Fish and Game Club

Appendix B Individual Interview Summaries

- comments in [] indicate information added by the interviewer to clarify what was meant by the interviewee

Very familiar with JR

*** Contact #1**

- worked at the powerhouse in the 70's fished at JR in the 60's
- fished steelhead approx. half a mile up stream (until area where you can't get past [boulder area])
- fish there in October, November
- uncertain of the number of steelhead
- steelhead and trout "of good size"
- fished before the mining accident - when cyanide dumped into the river
- afterwards (80's) cutthroat started coming back
- steelhead caught in large nets outside [JR]
- in Elliot there were no fish - reservoir was sterile
- a few cutthroat trout down Bear Creek
- a few trout in head pond of old power plant
- in the wooden flume the odd trout
- salmon spawned in the tail race
- all creeks in that area had good fish

*** Contact #2**

- lived in JR 1942 - 1955, visited the river afterwards
- his father was superintendent of the power plant
- Humpback and spring in the tail race, and up the JR as well
- spawned in the tailrace
- would see fish up until a mile up the river
- approx. 1 mile up there was a small pool that was salmon spawning bed
- trout from the mouth of JR to the dam site and into the reservoirs
- steelhead not so much in the river but some in the tailrace
- [I asked if he was aware of fish ladders on the river] no fish ladders in the canyon area - were fish ladders in other rivers - such as the river near Leechtown
- fish in the river in fall - September/October
- humpback 5-10 lbs.
- spring up to 20 lbs.
- cutthroat trout in both the tail race and JR
- bullhead in the river
- suggested contacting Allan Brooks (his dad was the fisheries officer-Jack Brooks) 642-5471, Soderbergs, Dick Poirier, Paul Michelson (in Sooke)

*** Contact #3**

- lived at Jordan River 1969-1973(74?) - before the new powerhouse was built
- Jordan River had: humpback/pink (blueback), coho, sockeye, jackspring, steelhead, cutthroat trout
- Steelhead in the river in late fall

- caught ocean-run cut throat trout off the logbooms (5-6 lbs.)
- Coho salmon caught below the powerhouse - early august - 10 lbs.
- fished in the area before boulders began, but had been past the boulders (drew a map of the area including spawning grounds)
- Provided a map and document regarding fishing on Jordan River
- Other possible contacts: Margaret Gifford (250) 384-0610 (mother); Laura Knight (250) 370-1079 (sister)

* Contact #4

- lived in JR approx. 1958-1970ish
- there were some spring (very few), some coho, huge run of pink (1000's), sea run cutthroat (could catch more than their limit), perch, smelt, steelhead (some)
- majority of pink in the tailrace
- thinks there is likely still arsenic and other chemicals in the streambed due to mine
- old cars were helicoptered into the river to plug up the hole at the mine
- steelhead went as far up as below the mine
- trout all the way up to camp 10 (bear creek dam)
- small trout
- fished for trout below camp 5 (Elliot dam) all the way down to JR
- very healthy trout (6 - 12 inches) because the water was running
- sea-run cutthroat in spring and fall
- steelhead 1-8 lbs.; spring - big; coho 5/6 lbs.; pinks 3-5 lbs.; sea-run cutthroat 2 lbs.
- perch 100 in each school - in the summer (July and August)
- suggested I contact Arnie Nyberg

* Contact #5

- grew up in JR 1949-1963
- in JR there were pinks, coho, chum (he thinks), steelhead, rainbow trout, cutthroat trout, herring, smelt, bullheads
- rainbow all the way down from the reservoirs to near the river mouth (weren't steelhead)
- rainbow in the "canyon area"
- the salmon spawned in the pool before the mine porthole (there was a waterfall up the river a bit which they couldn't get over)
- also spawned in the tailrace
- would also see the cutthroat and rainbow in the tailrace and the pool before the mine
- Time of year: perch, smelt and herring in the summer (June, July)
- salmon arrived in September
- steelhead in march or April until May and June
- cutthroat and rainbow year round
- Size: rainbow trout approx. 8-15 inches
- steelhead 8 lbs. was about average
- coho approx. 8 lbs.
- pinks approx. 4 or 5 lbs.
- cutthroat 12-14 inches
- Quantity: Coho approx. 500-1000; Pinks 1000+; Steelhead "plentiful"
- [asked if he was aware of any fish ladders on JR] - there were never any fish ladders on JR
- because of the boulders and fast running water it was too difficult for fish to get up
- suggested David Lionas (478-2480), William Lionas (658-6779), Dan Wade (478-1074) (also has brothers Arnold and Alan)

* Contact #6

- name from Glen Varney (Sooke Salmon Enhancement)
- in JR in 1950's, grew up there
- humpbacks(pinks) found in the spillways for the electrical station
- he was there in August to September (summer holidays) - humpbacks there at that time
- 2-4 lbs. - in the spawning process, mature
- trout fished as a kid - the resident rainbow
- trout fished above the boulder area
- in his teens the mine was active - steelhead fished at Muir and Kurby Creek
- wasn't at JR when the fish were spawning, he was playing in the creeks nearer to where he lived
- suggested I contact Les Poirier; Pat Forest; Teddy Elliot (father was superintendent of the powerhouse)

* Contact #7

- lived at JR 1943 - 1956
- spring, chum (pink), steelhead, cutthroat, coho, creek trout, perch
- on the river late Aug. to mid October
- coho late August/early September to mid October
- mouth of the river to 5 miles up there were fish ladders
- salmon on the river from the mouth to approx. 3 or 4 miles up
- steelhead up the river approx. 3 to as much as 5 miles up
- steelhead into the canyon area, had ladders in the area
- trout and cutthroat up the river into the lakes (up to camp 10)
- I asked specifically if there were salmon and/or steelhead in the "canyon" - said yes
- wasn't allowed to fish in the canyon area because of the variable flow due to the dams
- he could fish up to Camp 2
- coho 5-7 lbs.; chums/pinks 15-20 lbs.; steelhead 20 lbs.; cutthroat 2 lbs.
- lots of fish
- used to have fish in tailrace - coho, spring, chum, the odd cutthroat or steelhead
- tailrace would fill right up "you could walk on the fish"
- JR used to have several other canyons, now only little canyons
- remembers his dad talking about catching coho
- suggested that we contact his brothers as they have fished the river

Follow-up interview with Contact #5 on January 7, 2001

- There were no fish ladders in the river [this differs from his earlier comment that there were fish ladders]
- He was pretty young at the time (around 12 years of age)
- There were steelhead up to Camp 2, just below the dam
- There were salmon (spring, chum, coho and pink) 4-5 miles up the river, up as far as the first dam which is at Camp Five, about 4-5 miles up. Elliot dam came later, so the salmon were up past where this dam is now.
- Cutthroat up to Camp 10.

* Contact #8

Note that a MOE staff person conducted this interview, and that the interview was in person, and followed a different format than that used in the interviews conducted by RRL. The interviewer notes that the contact's recall was clear and accurate.

Several pertinent points can be made:

- His fishing experience relates to the reach upstream of the "obstruction".

- He claims to have caught and harvested steelhead in the late 1950's from the Elliot Reach. No pictures. [Why would anyone bother to fish the treacherous canyon if there were no substantive fish there to harvest]?
- He recalls fishing during March-April. 4. Typical steelhead catch was 2-3 fish per trip.
- He referred to a critical access tool to fish the canyon section, a wooden ladder. The ladder's location and access was adjacent to the Forebay Reservoir in Reach 2 with access from the East side.

* Contact #9

- fished on Jordan River in 40's and 50's - before the mine was put into effect (afterwards there were no fish- mining wiped out the fish in the 70's)
- in tail race from the old power house were coho and mostly pinks, some cutthroat
- size 3 lbs. - 8 lbs.
- a couple thousand coho and pinks
- fish there in late September - October
- returned every 2 or 3 years
- cutthroat trout caught out of the river
- other people caught steelhead up the river, just below the boulder area (he didn't catch it himself)
- didn't remember any of the fish being in the canyon area

* Contact #10

- grew up in JR in 50's and 60's, fished as a kid
- there were fish there until the mine backfilled (the logging didn't help the fish either)
- there were trout all the way up the river - small
- steelhead and cutthroat as far as the old mine
- would catch fish approx. 4 lbs.
- said approx. 100 caught each year (wouldn't catch one every time they went down there, but fished very often as a kid)
- fished as a kid with about six other guys
- Salmon spawning run up the tailrun
- humpback and/or coho (not sure)
- tailrace entirely full of fish - 100's if not 1000's
- smelt and perch every year
- 1000's
- [he] never went past the log dump (same place as the dry-land sort)
- bulltrout were present
- fished trout in the lakes - Diversion and Bear Creek
- steelhead never got up into the canyon - there were small trout in the canyon
- suggested Dick Poirier; Jack Brooks of Wiffen Spit Rd (retired fisheries officer in Sooke)

* Contact #11

- lived out there in 1964ish-1975
- -huge spawning area in tailrace - salmon (doesn't remember what species)
- tons of salmon "so many you could walk on them"
- used to be lots of perch
- cutthroat trout

- bull heads
- rainbow trout in "pools up higher"
- trout up further in JR, before the power project
- no steelhead in the river that he can remember
- suggested Dick Poirer; possibly Lajeunesses- the boys are slightly younger than he is

* Contact #12

- got his name from Cliff Banner
- fished steelhead in the winter 1954 - approx. 1965 (when the mine flooded and killed all the fish)
- fished at the mouth of the river in the salt water
- weren't big fish - less than 10 lbs., most approx. 6 lbs.
- not lots of fish, 3 or 4 regular fishermen visited the river
- fish only migrated to the waterfall
- had Chum in the tailrace of the power house

Familiar with the Jordan River but not much detail provided

* Contact #13

- worked in the hydro power house starting in 1952
- salmon run used to come up to the powerhouse tailrace
- tailrace used to be full of humpies [humpbacks or pink salmon]
- Suggested contacting Shakey [Les] Wade, Bob Lajeuness and Dick Poirer

* Contact #14

- there were small trout in the reservoirs, prior to the logging, in the 1970's
- lots of fish - had to check for limits
- the mine killed all the fish in the river
- didn't personally see a steelhead (Cold Creek, Muir and Tugwall all had good steelhead runs)
- suggested I try George Pednault

* Contact #15

- the Sooke Nation is involved in the consultative committee with BC Hydro
- have a particular interest in the area
- through BC Hydro's Aboriginal Affairs section plan to have an elders meeting with Sooke Nation elders to get historical information on the area (Pachenaht elders not included - the Pachenaht have not been attending the consultative committee meetings)
- suggested WF Products (Doug Cop); John Stevens 642-4756 (old fisheries officer)

* Contact #16

- weren't any fish in the Jordan River in 60's and 70's (that he was aware of)
- Jordan River wasn't where he would go fishing
- suggested to contact Stewart Soderberg - owner of Point no Point resort

* Contact #17

- name from Ted Burns
- ask about Frank Elliot (passed away)

- JR used to have Pink Salmon Run, prior to the when the mine broke
- not very familiar with the river, more familiar with the San Juan Basin

* Contact #18

- suggested several contacts including: George Reid at Ministry of Environment, Craig Whiteman and MELP, Victoria Fish and Game Oldtimers, Frank Elliot in Port Renfrew, Rob Waters, fisheries columnist for Times Colonist newspaper.

* Contact #19

- not familiar with the river
- has fished in one of the reservoirs when on his way to Wye Lake (resident trout)
- recreational fishing only, didn't make any notes
- thought that the Jordan may be a better stream for fish nor that the industrial impacts have decreased
- felt that it didn't have much recreation use because of the history of mine contamination
- felt that 2km was a good length for fish to use
- suggested going there, walking up to the barriers to see them
- suggested asking at the Jordan River Hotel

Connections

* Contact #20

- passed away
- his widow suggested Dave Slader who works for Hydro(?) and [lives(?)] in Sooke - [can find a number in Victoria but none in Sooke]

* Contact #21

- suggested I call Sooke Regional Museum

* Contact #22

- suggested I contact Jack Lenfesty (think 652-2495); Craig Wightman fisheries Bio at Nanaimo

* Contact #23

- Frank Elliot passed away (was knowledgeable of the area) - friend of Elliot
- suggested I contact Les Wade (previous owner of Shakies Hamburger stall)
- suggested Norm Smith (Port Renfrew marina in PR); Bob Galligaugher (Galligaugher's fish camp in PR)- to be familiar with the Gordon river not Jordan river

* Contact #24

- didn't have any information on the area
- suggested we speak with Port Renfrew

* Contact #25

- had already passed the names she had onto Hydro

* Contact #25

- President of Sooke Salmon Enhancement
- on the BC Hydro council looking at Jordan River; currently doing the Jordan River Fish Count

- suggested we contact Mr. Pat Forest (642-3988); Mr. Les Eves (642-5969); Mr. Mell Hull (642-5548); and Mr. Bill Pednault (642-3443)

* Contact #26

- didn't know information about the JR
- suggested I contact Randy Michelson of Sewar Rd- he grew up in JR

* Contact #27

- suggested Sooke Salmon Enhancement Society; Richard Poirier; Western Forest Products

* Contact #28

- retired DFO fisheries officer (1979 onward)
- JR used to have pink, steelhead and chum (info from written records)

* Contact #29

- suggested I speak with Wayne Zaccarelli - in charge of their fish section

* Contact #30

- didn't know of anyone off hand who had fished in the JR
- would take a couple of months to get a hold of members through their newsletter

* Contact #31

- Don Hall is their fisheries specialist
- not with in their area, suggested I call Helen Dunn from Pachenaht

* Contact #32

- not really familiar with the JR
- suggested Ron Ptolemy; Helen Dunn or Jeff Jones at Pachenaht

*Contacts #33

provided contact with the list of questions that we are asking -what species of fish (prior to 1970's); where on the river; what time of year; what years (every year?); quantity of fish; size of the fish (different species). If they hadn't mentioned it already I then asked - whether they knew of any steelhead on the river; where on the river; were they familiar with the canyon region, above the waterfalls - and were there any fish in that region, if so what types?

- she is contacting the Sooke Nation and they are planning to have an elders meeting in the next week or 10 days
- suggested I could also try Paul Sebar of the Dididaht (745-3333), he may have information re JR fishing
- I also mentioned that I am contacting Pachenaht and Nuu-chan-nulth TC - she hasn't been in contact with either

*Contacts #34-38

These individuals and organizations were contacted but did not respond.

Appendix C Jordan River Meeting with T'Sou-ke Elders

BC Hydro Water Use Plan

Jordan River Meeting with T'Sou-ke Elder's

December 11, 2000

Attendance:

Denise Purcell
David Lightly
Jim Cooper
Jack Planes
Frank Planes
Anne Wilson
Janie Hutchings
Michelle Sprinkling (Minutes)

Start Time: 10:15

David: There's a process going on that Hydro has initiated at the request of the government to look at all their facilities to re-access the use of the water. From hydro's point of view it's strictly the use of the water, a water use plan. We all came with naive little biologist saying lets look at the whole system and they said no, I am not digressing this has a point, what we want to do is look at the whole Jordan river and say what is the current state of it and what was it when you guys were young. Is there anything we can do to improve it to understand the Jordan river we have to know what has happened to it, so that's where you guys are really critical you cause you where there you watched it happen from the time you were young you were using the Jordan River, working there, fishing there so I think its really important that we capture that cause are ultimate goal is to look at the possibility of rehabilitation if that's impossible some kind of litigation.

Janie: I think just from our prospective in the water use plan it's a public consultation process so we have a number of government agencies in addition to BC Hydro so DFO is represented in the discussions as is MELP, there is also some individuals from the Sooke Community and the Jordan River Community so there is a number of things we've looked at, and people participating at that table have expressed what there interest are in how we supplied are uses of water there. Fishing is obviously one of the biggest ones, but we have also had an interest in wildlife and we operate the water there has impact on, could we improve the wildlife. Recreation there is a very high surfing interest, how we operate the water there because the surfers surf right on the river there. What we don't have in a number of cases we don't have any, we have the current what people are interested in currently, we don't have necessarily a perspective on what it was like before what were some of the traditional uses the re. That's what our prospective here is to see what kinds of activities were undertaken how was that really used and how is it currently used and are the things in how Bc Hydro operates that system that could be changed to make that access more useful.

Anne: I don't have any thing to add specifically and I just want to say that I really appreciate the opportunity to come here and learning from you and hearing how it was how that's most useful as far as water use stands.

Frank: I guess as the oldest one in this Nation should give you my opinion, when I was a young teenager, and before that my elder's told me that there was a good run of salmon in that river a very good run and it was used quite often by all of us. There was a big pink run, secondly Coho, and chum salmon, steelhead, sea run cut-throat and mostly sea run cut-throat the big ones they were all in there, I don't have anything on spring salmon, maybe Jimmy or Jack it seems to me that s a gap in my knowledge for spring salmon well I can't say that I didn't see I seen a few in there but not a big run but then again that's after they had all died off eh. But we used to fish there right in the spill way by the power plant there, in my time they were the only fish left, there were a few pinks in there and steelhead we used to fish steelhead in the spill way. I guess any one who fished in Jordan River would say the same thing. It is a big river and it was very important for the T'Sou-ke Nation because of its run of the four species that were really abundant in that river before the power plant went in I guess Jack is the one who worked there the longest he can take over for me know I told you all the important things like it used to be before the power plant went in and when the power plant went in like I say we used to fish

David: When would you have used, when would that have been when do you remember

Frank: well I was born in 23 and uh I guess I was what ten twelve years old when I was fishing steelhead and trout so you could take it from there. Jack can take over he worked there right from almost day one.

David: mid 30's

Jack: Jim do you have anything, you can go first

Jim: Yes

David: I have one more question for you Frank how about the falls what did you ever fish steelhead in the canyon up above

Frank: all over, all down the river

David: how far up did you go

Frank as far as I wanted to walk and that's quite a ways up eh, but we fished the whole thing, the fish were so abundant in there you didn't have to go far because if you wanted fish to eat you didn't have to go far.

David; did you ever see steel head above the falls, maybe half a mile up or maybe a little further than a mile were the a bearer to steelhead or did they go further up

Frank: not that I know of they weren't a barrier as far as I know like I say there were no roads in there them days and travel was only by were you could walk we didn't have to go to far to get what we wanted and still they were so abundant that you didn't have to go you catch them within the first two hundred yards of the mouth of the river so like I say they were that abundant that you didn't have to

David: this was before the mine

Frank: oh yeah, when did the mine come in there

Jack: Late sixties I think

Jim: yeah

Frank: we used to hunt in there, we also hunted in there swamps in there right where that dry land sort is we used to hunt mallards we used to hunt ducks in there were lots in there

David: do you remember a guy named Red Dunaway

Frank: yeah

David: he claims that you know more about the Jordan River than any man living

Frank: like I say I fished there whenever I, my brother used to take me there he was quite a bit older, carry me there, eh we used to he loved to fish trout and catch salmon, he would put me on his back and I would sit on a log and watch him fish that's one of the that's were I got my interest in fly fishing I've been a fly fisherman all my life, trout fisherman, salmon gatherer whatever you want to call me, name it I been a fisher since I was 9 years old went I fishing with my step father when I was 9 years old my sister and I did in rivers inlet from there on we I have never I used to fish in the summer log in the winter so

David: did the pink salmon use the main stem of the Jordan River just before the canyon

Frank they used every drop of water in that river,

David: not only in the power house they were right in the

Frank: they were right in every where the run was so heavy that well they had to go everywhere

David: so there must have been quite a bit of gravel right at the bottom of the canyon there were

Frank: oh yeah it was all spawning, at certain times you got in there you could almost kick them out of the water.

David: when did that gravel disappear from there

Frank: Jack probably knows more about that than I do cause he, like I say him and Jim have worked there so they probably know when the gravel was taken out of there. I spent four years in world war 2, in the dirty thirties that kind of took me out of there that's the only four years that I missed around here

Denise: touch on oral history, our relatives told us that it was major fishing village there, like before contact

Frank it was a major, major river they had cabins there this is what you see when we were really young boys, real wild bucks we would jump in canoe and go up, well there were little shacks everywhere we used to stay there in places this whole coast was marked out for canoe travel, was the only way we had to travel, we had no cars or airplanes we traveled by canoe so they had what you call canoe pull outs, Jordan river best one of the biggest for people coming down our going up, this is what my elder's told me then that was village for them to stay in

Janie: because you had the estuary there and I guess its flat and there's lots of space

Frank: oh yeah there was a camp ground, like a little village where you could go to get your salmon, you could smoke them there or load your canoe and bring them home

Janie: would that have been a year round village, seasonal, based on fishing

Frank: based on fishing, fishing run the camp was there year round. If you wanted to go traveling in the summer you were welcome to stay in the camp. If you got caught in bad weather in the straits of Juan De Fuca you pulled in there, it was one of the pull out places because you could get right in and out of the weather completely and the were with all the live to survive you weren't pulling into a place where there was nothing and you had nothing to eat you always fish to pull out of the water, so you wouldn't starve to death we used to jump into canoe and the first place we would stop was Jordan River, we would

be going to Nit Nat or San Juan. Jim was one of them we used to hunt seals along there to, we'd pull in and out of there that's what it was used for village site and stuff like that

David: did anything else go on there other than fishing like was there any other things you could think of

Frank: deer, hunting deer I mean it was always there was lots of deer in them days but know there aren't any but deer, ducks, fish like there was everything in there and all you had to do was wait until low tide go outside and you had all the seafood you wanted it was a, you could call it a planet of plenty you didn't have to want for anything you had good fresh water you had everything so

Janie: what about trapping, any trapping as well

Frank: I don't know, there probably was, but my uncle had a line in the Bay here weather he went that far or not I am not sure maybe one of the other guys has heard about it I used to boom around in the bay here in this trap line here

Janie: what did you trap

Frank: coons and mink and otter. We used to use that for an otters hunter place to when I, in my mid teens I guess we hunted otter, coons and mink unmercifully right up until world war two, we hunted them for day and night and we never ever brought them down to endangered species you can't there just that type of animal that you can't do that to and seals, of course in them days there was a great seal hunting place because in the dirty thirty's a seal the bounty on seal was five dollars then went down to 2.50, but even in the mid thirty's even five dollars to bring home to help the family a big help and if you got two or three coons a day or night your family ate well that's the way all them things wen tin them days it was gathering and hunting, if you in the winter time short of money you would go here and dig a sack clams, hunt seals hunt mink we used to get 18 dollars for mink and 21 for otter, and cougars where the last time I hunted a cougar 25 dollars each, that was big money today 25 come out of watch pocket, them days it put flour, coffee and tea on the table every coon you brought home

Anne: would you have hunted them in up in Jordan, would you have hunted the otter and raccoon and minks up in the Jordan up in that area

Frank: oh yeah, all the rivers these where river otter were hunting and the ones that had big runs of salmon had the otter were in there plentiful although they were hard to get but we got them, so any one can add to that

Janie: Denise do you have a map

Denise: I'll see

Jim: I think we can touch on, Jack knows about the cemeteries so I won't get into that, I worked in river just after war mid forty's, there was quite an abundance of salmon then that were they dumped there logs in the river and we made up the log booms, walk on logs and tried to spear them. But it is a settlement area that a lot of the bands not only the Canadian or San Juan but the Americans tribes it was a well known halibut fishing area and even right now I hear that Elwha that's outside Port Angeles there are claiming that in their lands claims along with Makaw they claim they come to Jordan River to fish I guess that's where the imaginary line went, it don't do them no good now but anyway that is a point of interest

Janie: because they cam e here

Jim: still is good Halibut fishing area

Frank: even today

Jim: salmon, steelhead trout of course we love to fish trout, didn't bother much with salmon but they were there. There was artifacts trough the area, the logging companies covered up a lot, Jack get into more, it

was a good area in the westwind when the canoe came in they could always go to the other end of the land where the office is there's a bit of a bender there and it helps them to land there canoes so they could land there in both types of wind east or west didn't effect them much, as I recall there when bottom fell out of Jordan River was when that mining company, slag pile and I don't know how that let go Jack could probably fill you in better there, but it seems they had some poison used to separate the copper from the mine and all that when it decayed it slipped into river and then as I recall I worked here at log booms here at Whiffen Spit I can remember not been able to use Jordan River it flooded and filled up with the rock and the tailings from the mine so they had to haul from Jordan River to Sooke here and that goes back I don't recall what year that was fifty's Jack

Jack: Sixty's and Seventy's

Jim: we hauled here for a couple of years

Frank; but most of the salmon had disappeared by the fifties, if you went right to the tailrace you could get one or two but they were pretty well gone

David: gone in river

Frank: Oh Yeah

Jack: I seen them in there when I first started in sixty's,

Frank: did you, were there very many in their Jack

Jack: you could look in that one pool above the dump ground there 500 pinks in that one pool when I first started there, massive pinks salmon that wasn't counting the chums and the Coho going in the tail race up Elliot not Elliot the one right the old power dam what did they call that one, anyway they had tail race come out of it

David: the big cement building that came in on the east bank

Jack; that's were the Coho and the chums would go and steel head and cut throat the pinks would go in the Jordan as far as they could go

David: just where it started to get steep and boulders

Jack: yeah it wasn't to boulderery then it was gravel but I can remember at least 500 pinks I n that one first pool there and there was two or three pools up above that

David When did that gravel disappear

Jack: well the gravel I guess when the mine flooded I am not sure what year it was it plugged the whole of Jordan River eh four feet of gravel and they couldn't dump there anymore and they were dumping in the water, so then they had to start hauling there logs to Sooke I was loader there at the time loading logs in the bush, I retired from logging 45 years in logging industry 27 of them I worked in Jordan river, started at Alaska Pine and went to River and all the way down the line Western Forest it is now so it's four or five different companies changed hands while I was there I can tell you we took some good timber in there should have seen the size of some of the timber we took out of the valleys

Jim: point of interest pay checks used to come from Michigan

Jack: yeah by mail

Jim: I can tell you at times the mail system wasn't that great there were times we had to wait three weeks for our paychecks

Jack: that's when it was Renoir, one of the biggest companies in the States owned the rights to Jordan River

Janie: my dad worked for Rainier to over at Woodfibre for awhile

Jack: oh yeah did he

Jim: CPS, Canadian Puget Sound, when Americans

Jack: the mine come in there I guess in the sixty's and took quite a few loads of? (oar) out of there, they had to wash it, so they pumped out of Fournier or I guess one of those lakes up there they built the flume from the old dam into the Fournier and that's how they run there and that's how they run the old dam the old power house they had that tail race all the way from dam up on mile high where they made that no what did they call that whet into the lake at four bay and from there it went into the old dam. Then hydro started building the new dam at Elliot and they dug a tunnel through the underneath stanski there and down into four miles of tunnel I guess it was, went to new power house that's there now the existing one that's were they draw the power now, they built the Elliot dam, that's below the. The one on mile high what did they call that the main dam

Janie: diversion dam

Jack: yeah just below the, I am going way back now to when they first built that and had the flew running into the old power house

David: that came in there was it 1913 or 16, so it's been a long, long time

Jack: so when the mine flooded I guess that four bay lake or something crashed trough they tunneled in the broke through and all the water came through

David: form four bay

Jack: must have come from up that way someway I think they did say it was four bay because it wasn't very far away, you could see it from four bay on that mountain where they drilled up they put a stack that sucked the air they give the boys something to breath I guess, that tunneled right under that whole mountain there, so it was all an under ground process thing was done under the whole milling everything was done under ground so they had to wash the ore, they made an 8 inch, 12 inch pipe come from the mine went down into the bay by office foreman's house a big pipe right into that bay all the tailings from the mine would go into that bay and killed everything it killed the seafood the crabs and it made as you go by in any boat or canoe you could see the whole bay was brown, as this mine tailings coming in changed the whole system from the Jordan River to the office, fine gravelly bottom fine sand like the San Juan all of that came out washed on beach, and turned into sand, in the meantime it was Sinead what they used to clean this ore and it polluted killed, the whole, I guess 2 miles ether side of everything that was existing there seafood and all the stuff that we eat, that from the mine when it filled up not counting what it did to four of gravel in the river, the Jordan River then they took these old cars and dropping them down these big tunnel to try and stop the water to this day I guess there' hundreds of cars pushed down into that mine

David: where were they dropping them right

Jack: right above there wherever they went in, they tried to stop it from coming out you could see the old remnants of the cars where it did finally broke, broke through flooded river when they flooded the river they had to quit logging, until they decided what to do well they got to keep logging they were quite a few hundred guys working at Jordan River, so the started hauling to Whiffen Spit were there logging trucks and they had big trucks at the time so they cut bunks to eight foot instead of twelve foot, so the used the big trucks alright but they made smaller loads, I was loading at the time I was the one putting logs on the truck to haul to Sooke, two to three years they must have hauled to Sooke, in the meantime

they put a spar tree with a great big machine on up behind there dump now they had a great big drag line bucket almost the size of this house I guess with a big line dragging it back and fourth to drag gravel out of the river they had two d nine cats and pushed right from the bar all the way up the river and this drag line would pull all the gravel up and they had to gravel the mountain there I guess it was almost as high as the hundred foot tree four or five hundred long all that gravel eh the machines were pushing it up and the were cleaning the river so they could start logging again so finally they got that cut down and the foreman looked at that big pile o gravel and said gees what are we going to do with all that, in the meantime behind us was a great big marsh

Frank: our duck whole

Jack: our duck hole, it channels in it with the Salmon when spawning they would go in there and raise, them to live the ducks would go in, the herrings any kind of wildlife lived in there, them prickly I guess they were willow's what do you call that

Frank: crab apple

Jack: it's really thick in there and the ducks and everything would go underneath it I guess it was I don't know how many acres I think they quoted it and eleven or twelve acres, he said well were going to put the gravel there so he started getting these great big trucks and off this pile filling in this pond he knew what he was doing cause he was going to make it a dry land sort there for the rest of the years they logged, he said he was going to build golf course but he was fooling the boys, you could find out how many acres that dry land sort is, over nine or eleven acres because that's all that gravel that came out of river

David: when was that, what year was that

Jack: well in the sixties

Jim: yeah in that area I guess

Jack: so that's the story about the mine when it broke trough and what else was there I forget what happened over the years

David: what were the flow like in the main Jordan River

Jack: well from the campground down there was no water just boulders when they blocked it off then when the big runs they had to put the spill way open up water would come in that's before Elliot was there before they built Elliot that was the main dam was at camp five they called it that big dam there now

David: yep diversion

Jack: I think they spent millions of dollars to make that earthquake proof or something a few years back.

Janie: yeah it is an ongoing process to keep our dams

Frank: excuse me, I have to leave

Jack: so when they started building the Elliot and I tell yea they build that tunnel in the one mine, they put a mining camp there above the flat on that side of the river, all these miners that come in to build tunnel there was so much people there working at Jordan River the logging going, mining going I think there was over 500 employed there they even had to put a bank in the hotel for all the people working there, CIBC moved a branch in there, lot of people working there they brought one of the biggest shovel in from Vancouver I guess to dig up there because all that tunnel rock coming out of the tunnel and they brought this barge in I had to go out in the tug to bring and they couldn't bring it across the bridge so we had to land it on the upper side of the bridge great big shovel ten yard bucket or something, one of the

biggest crawler shovels made so we took off the barge and walked it up hill, all of these tunnel rock that what they called it came out of this tunnel, two mining companies they worked together and they said when they broke through they were half way each, and the engineers where one inch off, when the tunnel broke through that's how good they were the tunnel I guess was a ten, twelve foot tunnel, through the solid rock I think there was four miles of it or something, you people might know more about that I am just going to what I

Janie: kilometers my self

Jack: whatever that same mountain that we logged Elliot comes right down to the new power dam, so we can go down to the new power dam when it moved in I was working there then I was on the boom, I guess they were having problems with it when it first started, all the little salmon and the good place Jordan River always had fish it had smelts, hooligans, shrimp I looked at bottom one day and the y whole bottom was moving I was sitting there waiting with a boom and I looked and I scooped them up and there was minute little shrimp and the whole bottom was moving I guess they go in there to grow up eh, always see fingerlings and all the baby salmon in the meantime they were still filling in the dry land slew that I was talking about

Anne: is that where you would see the fingerling salmon

Jack: all trough main

Anne: trough main river

Jack: wherever the creeks run out of it, its tidal there to its tidal water right up until the bridge there but there always so much life in that river, you could just see it you would just have to look out, the smelts we used to get a bucket full of smelts at night to take home I seen so many oulagans there once the seagulls couldn't even fly, the tide went out and they were all trapped in this great big tidal pool seagulls ate until they couldn't even fly that shows you what was there, there were oulagans we couldn't figure out what they were until we walked out and looked at hem four or five ton locked in tidal pool. But anyway when they got the tunnel trough new dam built and it was having problems and it had spill, Oil spill fine hydraulic oil, and I think they lost thousands gallons and they never let anybody know what was happening, nobody said word, we could see it the slick in the water fine hydraulic oil out of that brand new dam after that fish were gone killed it killed that river dead, after they cleaned the river out of this gravel they put the dam in, so many things killing river everything the hydro, the logging, the mine three big well we all made our living there we were drawing pay checks off it too so it had to be you couldn't stop it that's the way it was, but that's what happened to the river it died the mine killed it, the logging killed it I can hear stories before I started logging there, they used to have a flew that's how they go the logs off the bank into the Jordan River when they first started logging, run the logs off a they had to shoot, in the twenties I guess before the steam donkeys were there they had steam operated logging equipment there, I think the steam was just gone out by the time we started logging there Jim, big diesel engines then I think the last steam pot was on steam pot hill on the east side of the Jordan there the old steam pot sat there

David: probably still there

Jack: still there

Jim: point of interest after second world war all they camouflaged all there trucks Jordan River seen fit to buy a whole bunch of these logging trucks they didn't bother to paint so they still had there camo on and up and down the road these trucks

Jack: so that's how I seen Jordan River dye in the years I've been there it just died there was no more ducks no more fish no more shrimp no more oulagans no more smelts Coho's, chums pink salmon by the thousands when I first went there not counting all the other things that go along with it the marmots, otters and minks that eat all these things the wildlife that you see there and I've seen so many thing

happen in the years that I worked there. Jordan River we Jim and Frank talked about this that's are fishing station where we go to gather are food are main fishing station we all went there in canoes for halibut, salmon you heard it that's what it was the logging come in, the mine come in, the hydro come in I don't know how many years the hydro's been there the early twenties I guess

Anne: early teens I think

Jack: so there all a fault not hydro but the logging and mining all three combined at fault for killing Jordan River. Talk about the Burial ground now when I was working there the all knew I was native from Sooke and they new that was our territory especially the foreman not the foreman the superintendent oh I gone to fool that god dam Indian this time, every time he knew what I knew he tried to hide it because he didn't want me to see what they were doing, we were up behind where they had the logs up and the big all standing timber with spruce (flipping tape), look a this I found something I want to show you, coffins off the limbs of the trees still there you could the old boards and stuff buried them up in limbs so the animals wouldn't get them I guess holy Jesus that's something you have to watch that protest that well that's why I am telling you nobody knows it because its hard to find eh so I guess the superintendent found out that I knew about this and talked to Bob he was foreman I doubt he was foreman then he was still driving truck I told Bob I know that's there so every year that I worked there we had month holiday, so I took my month holiday, and when I come back that whole area was clear-cut they fell it they took the trees down that um

Jim: the burial trees

Jack: yeah they were logged and hauled out and he did that when I was on holiday, because he knew and I came back what the hell happened to all them trees, fooled Indian that time, he knew exactly what he was doing nothing I could do about it I knew it seen it I didn't take pictures of it and another instance we were logging up the Jordan River on the Sombero you could see it it slops, you could see the Juan de Fuca down there, called me on the phone come up here I want to show you something I guess the fallers there was an old big cedar log all bark off, I guess the fallers had to cut it in half cause it was so long, road crew going through clearing the road, Bob come and got me come and look at this, and it wasn't felled with a saw or an axe, it was felled with you could tell it must have been some kind of whatever they used to fall a trees in them days

David: Chisel

Jack yeah, I guess it wasn't good enough for canoe so they just left it and went to another one and he said well what should I do with it I said well he's got it bucked in half I'll get the loader and bring it down and put it in dryland there for the museum you could tell it wasn't modern day saw that cut it was done before even the white man come here I guess the used to build there canoes and drag them down put into the water after they built the canoes they would bring it out so they bucked this big cedar log he loaded on I told him load he said well I'll let you know what truck out two different trucks I was unloading down at the dryland so I picked them up and put them aside I got a picture of that somewhere we were going to bring them to the museum to Elida, summertime again on holidays you watch them two logs there that's going to the museum Bob said yep we'll look after it and I come back and where the hell are my cedar logs well we let the shake cutters in here and when you were shut down and they bucked all the cedar up there they cut my canoe log it was good clear cedar but I do have a picture I'll have to look through my pile of pictures somewhere and I got a picture of it that's the story of how they took the cedar out the bushes the canoes it was above the Somberio River, so that's the history how the natives got there canoe cedars from that area nothing we could do about it log was gone cut into shakes, all the years I worked there seen all that how they clear cut the most pristine places creeks and devastated it drug the logs across and plugged rivers and streams, the water would run off there would be big log jams would have to go there with cherry picker pull out these logs make the river flow again, road building, pushing in a road, in them days they pushed so far they would go in and load, they called it cherry picking build a right of way I was one unloader that would go and unload these logs that they felled go in the ravine with cat, and the big dump trucks come in and they dumped all these 45 gallons barrels off the road next thing buried with cat I said what the hell are you guys doing you didn't see any

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Denise: but you don't recollect steelhead

Jack: oh yeah

Denise: I mean like afterwards

Jack: no not after everyone one of these creeks had steelhead, that's my story about Jordan River it's our territory it's in our traditional territory for the T'Sou-ke People Questoo from Pacheedaht he's are uncle told us were to hunt and fish and who's territory we could hunt and fish whenever we want it's shared territory so he taught us a lot to hunt, fish catch halibut where and when remember he told me you put here not on paper you learn what I say I give you the land marks where to look, what time to go when the tide is right the color of the trees the fall the maples everything snag of the mountain were to fish in the straits everything don't put it on paper just hand it down to your kids so that's what I am doing

Janie: so you still using the Jordan River

Jack: oh yeah every chance we get not much these days if I want to go back I wan to see it like it was or I would like at least for our kids to see it, I would like to see some pinks back in there that's a natural river for making the must be like chums because they don't need that much area that Jordan River didn't seem that big like that canyon it only goes that far to have that many pinks in that river it must have been really good cause I said I seen at least five hundred in that pool at least

David: what was the gravel like just right where they spawned

Jack: it was good gravel just like you see that (milne?) pool over there same kind of gravel until it all washed out turned into boulders that's all that was left great big boulders no more at all

Janie: so when the pinks where in the river and the Coho and the chum were in the old tail race that was prior to new dam, and then after the new dam came in

Jack: after new dam the coho's were still trying to get into the new dam where the little flews coming out there we used to go and see the little cohos going right up into to where the water is spilling out right up until maybe fifteen years ago I guess probably still a few Coho go in there now I don't know have you spotted any Coho in there at all

David: yah I thin there was three

Jack: that's showing you that they are trying

David: yeah they stray from one system to another

Jack: yeah I seen them right up until I retired out of there but I guess the biggest thing was that Sinead washed away tailings from mine eh just killed the whole area and it turned that beach into sand but it's all washed away now since that pipe come out you know tailings from that mine

David: one thing piled on the other, that's what I was trying to get at that statement at the begging is that its really hard to sort out what are you going to do with the river, hydro has one interest and that's is there anything you can do with current hydro station there to make it better for fish, but there so many things that piled on top of it that its really a what are we going to do with it

Jack; hydro's only interest was putting the power getting the dam powered and get power for Victoria I guess they said it was a back up when they couldn't get power anywhere lese which it was the got miles and miles of water backed up back there all the way past bear creek and all that miles of water behind the Elliot dam now

David: There running a bunch of models hydro engineers and the first thing they told us was everything's full and you turn on the tap and generate maximum power your out of water in three days, it's a huge generator

Jack: its just for back up, we logged that area and up yea know because they had to flood it in the old dam the trees were still there we logged it with a machine then we had to pull the limbs and everything off all they left was the stumps because they flooded that

David: so what are we asking hydro what would it be like if you gave some water for fish and its more complicated than this the bottom line is you either have power or water to support fish with something like there natural state so

Jack: well with the Elliot in there now they should have lot water in there for fish

Anne: The Elliot's only got a small capacity for holding water so that the problem in terms of the supply of water its just a small reservoir

Jack: you can almost watch the Elliot go down when they turn the power on in the big rains it used to spill one or two feet over the top of the Elliot it couldn't hold that much water back in the big rains up in the hills when we had big rains I would see Elliot spill two feet over the top and he would call the guy at Jordan River office say that the dam is starting to spill we have to quit booming or get the logs out of there cause she's starting spill so we had to phone the tug and get the boom and let the river go down so they had to get communication between the guy at the dam and the office come and phone me I was at the boom at the time Elliots starting to spill now you got to quit dumping

David: what would you guys like to see or what would you like the Jordan River to look like

Jack: I would like to see some more humpy's some more fish, hit the lakes put them in there see what happens

Jim: my thoughts are we don't have to go in time as I recall the early fifties it wasn't that bad surely they must see or must no that when you wreck something in the environment this day and age your obliged to not just some cosmetic little thing and say ok that isn't good enough like borrowing your car if I wreck it I am obliged to put it back together so I think that we talk in term s of abundance sure there was an abundance but all the rivers up and down the island are some depleted, beside the Sooke and all the smaller ones, for them it would be a real feather in there hat to say look we the River Jordan, as opposed to just walking away from it them are the breaks would go a long way for them to there programs to try and do something about it and point the finger totally to hydro, the dam was the crusher, the logging and the mining had there little in discretion's, they blew away river, in term s of a logger less water comes out of there, for the purpose that hydro is using it for now they don't have no value other than what there using it for, my dad found a Spanish Muscatine bedded in a cedar tree when it takes a lighting stick it peels off one side of the tree exposes the bark and has a little bit of hollow in stump the Spaniards were up and down the land and all that the wood part was gone but the metal piece was still there

Jack: that was found back there wasn't it Jim, up in Valentine somewhere

Jim: Yeah it seems that its not a real if I was living in the Ivory Tower in Vancouver r look at it we can go a long way to building public situation her e by rebuilding the Jordan or to appoint or they can see that there doing something this probably isn't the first situation the have come across major rebuilding programs and uh not entirely there fault, but there the ones that are left to kind of clean up the mess they should they really should do something to the Jordan, dollars sometimes is just dollars it don't buy you nothing it don't give you any it don't buy you anything other than hydrates give them a couple three hundred thousand dollars and it would be all over with but to leave the Jordan the way it is what good is that seven hundred thousand if it hadn't gone back into it they have a my way of thinking is not uh, the best way to bring back the public image is take the guys from Sooke take the natives from Sooke and put them to work they have a lot of trust funds they give out for the publics we we've enjoyed some of

there money here for building our canoes a restoration project, FRBC in place weather they have, joint project to do something we have seen the best of Jordan River in our time, we had a lot of fun we took it for granted, the last culprit was the mine, it really did it all the tailings from that mine flooded into the river the boulders and it created a situation where it destroyed all the habitat for the fish no the begging of the end but the end I guess

David: what role do you think the Douglas Treaty has as a play in this

Jack: that's are hunting and fishing stations to begin with to fish and hunt as formerly so that still stands if we wanted to hunt and fish out there we still have rights it is a fishing station its an old fishing station registered for the T'Sou-ke territory so I don't know if anyone could stop us if they wanted but is there anything there to hunt and fish anymore

David: I was thinking about the extension if there is a treaty that says you can fish as formerly the habitat supporting fish has been destroyed where does that leave the treaty

Jack: well that's a violation of our treaty right then isn't it they took that anyway from us

Jim: the government is responsible for ensuring that these rights are protected and there's no question that the American tribes when they came over that the americium tribes, the American tribes, there's' probably battles that was a food source it was situation is here now, the Americans they came over and that was the invasion of the Jordan River the battle were fought we really have a blessed interest there some of our people spilled blood to protect to it

Jack: now we have future generations to look after so they should enjoy what's left there if they ever restore it try to restore it, we know the water use will continue and we know the hydro is going to continue we don't know if there going to put another mine if someone's going to start mining again

David: there is talk of reopening that mine so it would be a different story

Jack: better restrictions than they did last time I don't think they even had a permit to put that pipe down in there they put it out maybe 200 yards into bay and started putting the tailings there flooded that whole beach, beach destroyed all the sand and stuff they took out of there

Jim: so what is hydro, do they want to expand or what do they want to do there, they must have known the (?) all the years to come along now and say were interested in doing this we want to now about it that really don't wash with me there must be an interest there that they must have or want to do something want to rectify there sins or whatever the case maybe we don't now to come along fifty years after it happened thirty years after it happened and say we want to do something now

Janie: it might be an exaggeration to say we want to rectify our sins, we are in a situation though were we are doing water use planning and things like this through all our facilities there is a recognition that when we put in the facilities that there was no that it was a different time like when they did the mine, you didn't ask the people that lived in the area what there interests were you didn't, you weren't asked what impacts you were having on the environment some of where were at is we don't have any records of what the decision making process was for example the Jordan River Facility where were at now is working with the government and the other agencies to look at what are the interests of people now that are guiding how we make decisions about given that the system is there what are we who are we talking to what interests are we taking into account to look at how we operate the system in a way that's better for the fish better for the environment and still gives hydro some of what it wants power so its trying to balance some of that out and not just saying hydro wants power and were here and were going to do it we are talking to some of the other interests in particular fish and wildlife

Jack: bottom line though is that hydro has to be pushed

Jim: they want exclusive rights to water do I see something here that they got the water they want the water they got it and all were interested in is the water to make there generator work

Janie: as part of this process that's right were not looking at making any changes to the facilities how we operate the system, how much water we use when we use the water right now we operate the system in a way that produces the most power for hydro and the biggest buck for hydro, is there something but maybe you need more detail on that but for example we didn't know that there was an interest in surfing that was impacted by how we use the water so now we know that we are looking at is there for example can we turn the generators on at a different time of day so that it still benefits hydro but doesn't have a negative impact on the surfers and those same kinds of questions

Jack: your more interested in them surfers because they say that the water that s coming out of the river is impacting there surfing is that what you are telling me

Janie: that's one of the interests that's been expressed

Jack: or is that what they are telling you

Janie: well they came

Jack: that surf rolls in from the ocean on a big ground swell coming from the big seas outside, not from the river

Janie: we know they use the ocean coming in but when we release water it has some impacts on the quality of the surf

Jack: well there's other places to surf other than the river where the river is running out the surf from both sides, if there telling you that your ruining there surfing on account of that river there telling you lies.

Janie: I wouldn't say they are going to that extent

Jack: They way you talked there it sounded to me like they had you brain washed I know how them ground swells come in there they come in from the ocean because of the big seas outside and the big ground swell comes up its not the river doing it you'll see an impact from the river when its running sure you might get a bigger ground swell you know make it better but there going to surf there weather that rivers running or not there's going to be surf there and

Janie: I didn't mean to give the impression that surfing was becoming the primary interest in the process that's just one of the interests that's been addressed

Jack: I think that fish I should have a lot more priority than them god dam surfers I'll tell you that right now

Janie: I think your opinion would be echoed by others at the table

Jack: it sounded to me like there had you brainwashed half of them surfers don't even come from this area they come from all over the United States and Germany and everything else and they complain about our surf so you now what I am trying to tell you

Janie: yeah so don't come here

Jim: you don't have a son that's a surfer or anything

Janie: not that I am aware of

Jack: probably got a (?) somewhere in Germany saying that Jordan River has best surfing on the coast, so they get the priority and our fish and wildlife is at the bottom and that's not right

Janie: and that's not correct

Jack: I hope it isn't if they tell you that, that river is spoiling there surf then there telling you lies cause I ain't seen it I've worked there I put booms out of there and its not that river has nothing to do with it with the surf

Jim: surf comes off just before the high water flats when the tide ends the flat straight out that's the current unless there's a (?) then it pile s up the surf

Jack: no doubt there's room for everybody I have nothing against surfing but they want to put a board out in the ocean and surf that's there business don't start blaming the river you know

Anne: I think that Janie was just using that as an example but really the provincial government has said values have changed in terms of how power is operated and BC Hydro is a crown corporation so it is owned by the province and the province is basically saying look we generate power we've been primarily interested in using water for power generation only and going through what this process is going through and looking at each facilities and saying ok what are the other values in the area and make the choices explicit if your going to be generating power well then your doing at the determent of fish habitat or the determent of wildlife habits, so its been really clear about what the choices are and what and secondly the government is saying the values are changing so other none power benefits other benefits values do have value and so hydro's making some trade offs to say hey well take a power reduction in this area or for it and other values to so the so that's where were at in this process so right now it's sort of collecting the information what are the interests what are the values what did it look like what's possible being clear on if Jordan River generating this way then what are some possibilities benefits what are the trade offs between them and so like listening to you and getting information from you is extremely valuable and appreciated and knowing what were the species up there and what times and for instance steelhead there's a question of whether steelhead where above the falls and above the bearers and if you have recollection about fishing for steelhead up higher up in the river that's all of this information is really useful cause there is none anywhere else except for in your memory

Jack: well when that dam was shut off at camp five that Jordan river is just a trickle no water there at all until it's back flooding and they open it to let the water out then you got water but when there shut off there's not a once of water in that bloody river its just boulders the only little bit of water you get is the little through creeks you get on the way down other than that there is no water steelhead used to go there when it was natural they would go as far as the could get when they cut it off the water was shut off until they opened it up and that's control and they open when there's to much and shut it off when they need it

Anne: How far could they get, the steelhead

Jack: they must have got all the way to (?) cause that's where that dam was from the dam there was a flew so when they release the water in didn't go into the Jordan River it went in to a side channel that flew went into four bay lake to a generating plant from camp five where they blocked the water it come from a flew from this lake into a smaller dam and then to Jordan river here with nothing in it until it spills non fish no water you got no fish because ordinary dams when it gets full you generate the power the power goes back in the river but this has flume for five miles or something then into another lake and then into the old power house there's you r difference in the Jordan River, Jordan River had no water while dam there, so you had no more steelhead except when they went into the tidal water about half a mile or so up with the tide that's as far as they could get then because there was no more water in Jordan it was blocked off you can take it from there you had steel head until dam came in whatever year that was its as simple as that

David: we don't have much information on how the old system was operated the way it is now hydro tries to maximize the power output by putting most of the water through the Jordan

Jack: the only time Jordan got water is when they had an abundance and they opened the flume and that's that other power the old power house up and running the water was dumped on that side of the flume it didn't go into the river after they used it, used it came out of the tail ways out of the old dam and the salmon would go there not even I don't think that tail race was very long it was only half a mile long smell that water you knew it was the water that they used to spawn there eggs weather it was diverted around the corner or not water from that upper region you know

Jim: then the way I understand it hydro has no interest in withstanding any of there operations out there or like another power house or its nothing

Jack: Well they talked about raising the Elliot a few more feet didn't they

Anne: a few years ago they did because not raising it the Elliot non the Elliot sorry Bear Creek I guess the structure at Bear Creek it needed an upgrade like for a (?) in case of an earthquake so they cost it out weather it was worth doing the work and they decided against it that was the most recent look into that

David: huge generator it's really impressive to see it the shaft on its like this

Jack: yeah

Jim: which one in the Elliot dam

Jack: just above the water just above Jordan River there where the new dam went in that water coming out of that tunnel from the Elliot that's running that hydro its still generating power there

Anne: yeah

Janie: oh yeah

Jack: toured it a couple of times three in there

Anne: One big one

David: 170 mega watts

Anne: biggest one on the island

David: it's designed to turn on whenever anybody comes home and turns on there stove

Jack: you might not hear anything about that spill I am talking about but it was the finest kind of hydraulic oil

David: out of the

Jack: out of the new dam

Anne: that was the first time I've heard of it actually

Jack: we knew what happened and it was thousands gallons in that whole area you know when you poor oil on water it thick as a table the whole river

David: so that was out of the new powerhouse

Jack: it was just started they were

Janie: testing or something

Jack: and something broke and thousands gallons of that fine hydraulic oil went into that river I was booming there at the time I seen it

David: how long did it last

Jack: week or so until nature flushed it clean I guess oh no I guess it went into the beaches it disappeared, it wasn't black oil it was so plastic clear like that clear hydraulic oil

David: so you think that had a big effect on the river

Jack: I don't know what it did I couldn't say all I know is you put oil in the water you going to kill something must have cost them thousands of dollars I don't know how many gallons it was nobody probably even heard about it, a few other people around heard about it

Denise came in with lunch

Jim: well when you got an interest means a lot, you can understand raise our voices it's not that you wrecked it personally but you're the closest we can vent our anger to.

Jack: over the years you see it happen and wonder what's really happening how can you correct it you know the damage is done

Tape ended meeting over for lunch at 11:55am